

Reproduction

Human Physiology - Chapter 20

Biology Teaching Resources

December 24, 2025

Outline

20.1 Sexual Reproduction

20.2 Endocrine Regulation

20.3 Male Reproductive System

20.4 Female Reproductive System

20.5 Menstrual Cycle

20.6 Fertilization, Pregnancy, and Parturition

Summary

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Summary

Sex Determination

- **Sexual reproduction** (有性生殖) involves fusion of **gametes** (配子) - sperm and ova
- **Germ cells** (生殖细胞) formed in **gonads** (性腺) through **meiosis** (减数分裂)
- Chromosome number reduced from 46 to 23 in gametes
- **Fertilization** (受精) restores diploid number (46) in **zygote** (受精卵)

Sex Determination

- The human life cycle alternates between haploid and diploid states
- Meiosis produces haploid gametes (23 chromosomes)
- Fertilization produces diploid zygote (46 chromosomes)
- This cycle maintains genetic stability across generations



Sex Determination

- **Chromosomal sex** (染色体性别) determined at conception
- Females: XX chromosomes
- Males: XY chromosomes
- Y chromosome contains **SRY gene** (性别决定区基因) encoding **TDF** (testis-determining factor, 睾丸决定因子)

Sex Determination

- The Y chromosome is much smaller than X chromosome
- Despite small size, Y chromosome carries critical genes for male development
- SRY gene is the master regulator of sex determination



Sex Determination

- **X-inactivation** (X 染色体失活) occurs in female cells
- One X chromosome becomes inactive forming **Barr body** (巴氏小体)
- This achieves **dosage compensation** (剂量补偿) between males and females
- Random inactivation leads to mosaic pattern in females



Sex Determination

- Early embryos have **indifferent gonads** (未分化性腺)
- Gonads can develop into either testes or ovaries
- Presence of TDF directs testis development (week 7)
- Without TDF, ovaries develop by default (week 9-10)



Development of Accessory Sex Organs

- Two duct systems present in early embryo:
 - **Wolffian ducts** (沃尔夫管) → male structures
 - **Müllerian ducts** (苗勒管) → female structures
- Hormones from gonads determine which system develops

Development of Accessory Sex Organs

- In males: testosterone stimulates Wolffian duct development
- **MIF** (müllerian inhibition factor, 苗勒管抑制因子) causes Müllerian duct regression
- In females: without testosterone and MIF, Müllerian ducts develop
- Wolffian ducts regress without testosterone



Development of Accessory Sex Organs

- External genitalia develop from common primordia
- Genital tubercle, urogenital folds, labioscrotal swellings
- Male development (with testosterone): tubercle → penis; folds fuse; swellings → scrotum
- Female development (without testosterone): tubercle → clitoris; folds → labia minora; swellings → labia majora



Development of Accessory Sex Organs

- Developmental timetable shows critical periods for sexual differentiation
- Sex determination → gonad development → duct system development → external genitalia
- Most differentiation complete by week 12 of development



Development of Accessory Sex Organs

- **DHT (dihydrotestosterone)** (双氢睾酮) is more potent than testosterone
- Enzyme **5 α -reductase** (5 α -还原酶) converts testosterone to DHT
- DHT essential for male external genitalia and prostate development



Disorders of Sexual Development

- **Androgen insensitivity syndrome (AIS)** (雄激素不敏感综合征): XY individuals with non-functional receptors develop female external genitalia
- **5 α -reductase deficiency** (5 α -还原酶缺乏症): XY individuals cannot make DHT, resulting in ambiguous genitalia
- **Congenital adrenal hyperplasia (CAH)** (先天性肾上腺增生): XX individuals exposed to excess androgens
- These reveal critical roles of hormones and receptors in development

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Summary

Steroid Hormone Biosynthesis

- **Steroid hormones** (类固醇激素)
derived from cholesterol
- Common biosynthetic pathway for all steroid hormones
- Testes and ovaries can produce both androgens and estrogens
- Differ in proportions produced



Hypothalamic-Pituitary-Gonadal Axis

- **HPG axis** (下丘脑-垂体-性腺轴)
regulates reproduction
- **GnRH** (gonadotropin-releasing hormone, 促性腺激素释放激素) from hypothalamus
- Stimulates anterior pituitary to secrete FSH and LH
- **FSH** (follicle-stimulating hormone, 促卵泡激素)
- **LH** (luteinizing hormone, 促黄体激素)



Hypothalamic-Pituitary-Gonadal Axis

- **Negative feedback** (负反馈) maintains hormonal homeostasis
- Sex steroids inhibit GnRH and gonadotropin secretion
- This feedback loop regulates reproductive function
- Balance between stimulation and inhibition

Puberty

- **Puberty** (青春期) initiated by increased GnRH secretion
- Occurs in pulsatile fashion
- Hypothalamic sensitivity to negative feedback decreases
- Allows increased gonadotropin secretion
- Triggers gonadal maturation and sex steroid production

Puberty

- Girls' puberty typically begins age 10-12
- **Thelarche** (乳房发育): breast development
- **Pubarche** (阴毛发育): pubic and axillary hair growth
- **Menarche** (初潮): first menstrual period
- Changes in body composition and fat distribution



Puberty

- Boys' puberty typically begins age 12-14
- Testicular and penile growth
- Voice deepening due to laryngeal growth
- Facial and body hair growth
- Increased muscle mass and bone density
- Onset of spermatogenesis



Puberty

- Growth spurt timing differs between sexes
- Females: growth spurt at age 11-12
- Males: growth spurt at age 13-14
- Girls temporarily taller than boys in early adolescence
- Males eventually achieve greater adult height



Other Regulatory Factors

- **Pineal gland** (松果体) secretes **melatonin** (褪黑素)
- Melatonin may inhibit GnRH and delay puberty
- Photoperiod influences reproductive seasonality in many animals
- **Libido** (性欲) influenced by testosterone in both sexes

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Summary

Testicular Structure

- **Testes** (睾丸) produce sperm and testosterone
- **Seminiferous tubules** (曲细精管): site of spermatogenesis
- **Interstitial (Leydig) cells** (间质细胞): secrete testosterone
- Spatial separation allows independent regulation



Hormonal Control

- FSH acts on **Sertoli cells** (支持细胞) to support spermatogenesis
- LH stimulates Leydig cells to secrete testosterone
- Testosterone provides negative feedback on LH
- **Inhibin** (抑制素) from Sertoli cells inhibits FSH



Hormonal Control

- Dual feedback system allows independent regulation
- Testosterone production can be regulated separately from spermatogenesis
- FSH and inhibin regulate sperm production
- LH and testosterone regulate androgen levels

Testosterone Metabolism

- Testosterone converted to more potent forms in target tissues
- **5 α -reductase** (5 α -还原酶) converts testosterone to DHT
- **Aromatase** (芳香化酶) converts testosterone to estradiol
- Different metabolites mediate different effects



Testosterone Metabolism

- Androgens have diverse physiological effects
- Development of reproductive organs during embryonic life
- Secondary sex characteristics at puberty
- Protein anabolism and muscle growth
- Bone growth and maintenance
- Behavioral and cognitive effects



Testosterone Metabolism

- Testosterone acts within testes via paracrine mechanisms
- Intratesticular concentration 100× higher than blood
- High local concentration essential for spermatogenesis
- Demonstrates importance of local hormone action



Spermatogenesis

- Spermatogenesis takes approximately 74 days
- **Spermatogonia** (精原细胞) are diploid stem cells
- Some remain as stem cells; others become primary spermatocytes
- Ensures continuous sperm production throughout life

Spermatogenesis

- **Primary spermatocytes** (初级精母细胞) undergo meiosis I
- Form **secondary spermatocytes** (次级精母细胞)
- Secondary spermatocytes undergo meiosis II
- Form haploid **spermatids** (精子细胞)



Spermatogenesis

- Spermatogenesis occurs in stages within germinal epithelium
- Sertoli cells provide support, nutrients, and regulation
- **Blood-testis barrier** (血睪屏障) protects developing sperm
- Tight junctions between Sertoli cells create immune privilege



Spermatogenesis

- **Spermiogenesis** (精子形成)
transforms spermatids into spermatozoa
- Acrosome formation over nucleus
- Nuclear condensation
- Mitochondrial arrangement in midpiece
- Tail development
- Excess cytoplasm discarded as residual bodies



Spermatogenesis

- Mature spermatozoon has three regions
- **Head**: nucleus with condensed DNA; **acrosome** (顶体) with enzymes
- **Midpiece**: mitochondria providing ATP
- **Tail**: flagellum for motility
- Highly specialized for swimming to and fertilizing ovum



Male Accessory Organs

- Male reproductive system includes testes, ducts, and glands
- **Epididymis** (附睪): sperm maturation and storage
- **Ductus deferens** (输精管): transports sperm
- **Seminal vesicles** (精囊): fructose, prostaglandins (60% semen)
- **Prostate** (前列腺): alkaline fluid (30% semen)
- **Bulbourethral glands** (尿道球腺): mucus for lubrication



Male Accessory Organs

- **Penis** (阴茎) contains erectile tissue
- Two **corpora cavernosa** (阴茎海绵体)
- One **corpus spongiosum** (尿道海绵体)
- Erectile tissue has vascular spaces that fill with blood



Erection Mechanism

- **Erection** (勃起) from parasympathetic stimulation
- **Nitric oxide (NO)** (一氧化氮) is key mediator
- NO activates guanylate cyclase → increases cGMP
- cGMP closes Ca^{2+} channels → smooth muscle relaxation
- Vasodilation allows blood to fill erectile tissue



Erection Mechanism

- **PDE5** (phosphodiesterase type 5, 5 型磷酸二酯酶) breaks down cGMP
- **Sildenafil (Viagra)** (西地那非) inhibits PDE5
- Prolongs cGMP action and enhances erection
- Treats **erectile dysfunction** (勃起功能障碍)

Erection Mechanism

- **Emission** (射出): movement of sperm/secretions to urethra (sympathetic)
- **Ejaculation** (射精): expulsion of semen through contractions
- **Semen** (精液): 100-400 million sperm per ejaculate
- Contains sperm plus glandular secretions

Male Fertility

- Normal semen parameters important for fertility
- Volume: 2-5 mL
- Sperm count: >20 million/mL
- Motility: >50% motile
- Morphology: >30% normal shape



Male Fertility

- **Vasectomy** (输精管切除术) is surgical sterilization
- Segment of ductus deferens removed or blocked
- Prevents sperm from entering semen
- Does not affect testosterone or sexual function



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Summary

Female Reproductive Anatomy

- **Ovaries** (卵巢): produce ova and hormones
- **Uterine tubes** (输卵管): transport ova; site of fertilization
- **Uterus** (子宫): environment for embryo and fetus
- **Cervix** (子宫颈): neck of uterus
- **Vagina** (阴道): canal to external environment



Female Reproductive Anatomy

- Sagittal view shows anatomical relationships
- Reproductive organs positioned between bladder and rectum
- Uterus normally anteverted (tilted forward)
- Understanding anatomy essential for clinical practice



Female Reproductive Anatomy

- External genitalia (**vulva**, 外阴) includes:
 - **Labia majora** (大阴唇): homologous to scrotum
 - **Labia minora** (小阴唇): smaller inner folds
 - **Clitoris** (阴蒂): homologous to penis
- Demonstrates homology between male and female structures



Ovarian Follicle Development

- Ovaries contain follicles at various developmental stages
- **Primordial follicles** (原始卵泡): primary oocyte arrested in prophase I
- **Primary follicles** (初级卵泡): oocyte with granulosa cells
- **Secondary follicles** (次级卵泡): fluid-filled spaces develop
- **Graafian follicle** (格拉夫卵泡): mature follicle with large antrum



Ovarian Follicle Development

- Primary oocyte resumes meiosis just before ovulation
- Completes meiosis I → secondary oocyte + first polar body
- Secondary oocyte arrested in metaphase II
- Meiosis II completed only if fertilization occurs



Ovarian Follicle Development

- Ovary contains follicles at different stages simultaneously
- Growing follicles, mature follicle, corpus luteum
- **Atretic follicles** (闭锁卵泡): dying by apoptosis
- **Corpus albicans** (白体): scar tissue from degenerated corpus luteum



Ovulation

- **Ovulation** (排卵) is release of secondary oocyte
- Triggered by LH surge at mid-cycle
- Oocyte released with surrounding corona radiata
- Cloud of follicular fluid accompanies release



Ovulation

- **Oogenesis** (卵子发生) differs from spermatogenesis
- Begins during fetal development
- Primary oocytes arrested until puberty
- Monthly resumption: meiosis I → secondary oocyte + first polar body
- Meiosis II completed at fertilization → ovum + second polar body



Ovulation

- Monthly ovarian cycle progresses through stages
- Primordial → primary → secondary → graafian follicle
- Ovulation releases oocyte
- Corpus luteum forms from ruptured follicle
- Degenerates to corpus albicans if no pregnancy



Hormonal Regulation

- FSH stimulates follicle growth and estrogen secretion
- LH stimulates theca cells to produce androgens
- Granulosa cells convert androgens to estrogen
- Rising estrogen initially inhibits FSH/LH (negative feedback)
- High estrogen triggers LH surge (positive feedback) → ovulation

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Hormonal Changes

- **Menstrual cycle** (月经周期) averages 28 days
- Coordinated changes in FSH, LH, estrogen, progesterone
- LH surge at mid-cycle triggers ovulation
- Hormonal patterns ensure endometrial preparation for implantation



Hormonal Changes

- Integration of pituitary, ovarian, and endometrial cycles
- Pituitary hormones drive ovarian changes
- Ovarian hormones regulate endometrial changes
- Downward arrows show hormonal effects



Ovarian Cycle Phases

- **Follicular phase** (卵泡期): Days 1-14
 - FSH stimulates follicle development
 - Rising estrogen from growing follicles
 - Ends with LH surge and ovulation
- **Luteal phase** (黄体期): Days 14-28
 - Corpus luteum secretes progesterone and estrogen
 - If no pregnancy, corpus luteum degenerates
 - Hormone drop triggers menstruation

Ovarian Cycle Phases

- **Corpus luteum** (黄体) is yellowish endocrine structure
- Forms from ruptured follicle after ovulation
- Secretes progesterone and estrogen
- Essential for maintaining early pregnancy



Ovarian Cycle Phases

- Endocrine control involves feedback loops
- Negative feedback dominates most of cycle
- Positive feedback triggers LH surge at mid-cycle
- Integration of hypothalamic, pituitary, and ovarian signals



Ovarian Cycle Phases

- Menstrual cycle phases clearly defined
- Each phase characterized by specific hormonal patterns
- Endometrial changes correspond to ovarian events
- Understanding phases essential for fertility management



Endometrial Changes

- **Endometrium** (子宫内膜) undergoes cyclic changes
- **Menstrual phase** (月经期): Days 1-5, shedding of functional layer
- **Proliferative phase** (增殖期): Days 5-14, estrogen-stimulated growth
- **Secretory phase** (分泌期): Days 14-28, progesterone-stimulated secretion
- Without implantation, progesterone withdrawal causes menstruation

Temperature and Fertility

- **Basal body temperature** (基础体温) rises after ovulation
- Increase of $0.3\text{-}0.5^{\circ}\text{C}$ due to progesterone
- Can be used for fertility awareness
- Helps identify ovulation timing for natural family planning



Factors Affecting Cycles

- **Pheromones** (信息素) may synchronize cycles among women
- Stress disrupts cycles through effects on GnRH
- Body fat affects reproduction:
 - Very low fat (<17%) can cause **amenorrhea** (闭经)
 - Obesity disrupts cycles through hormone metabolism
- Minimum body fat required for normal cycles

Contraception and Menopause

- Contraceptive methods include:
 - Oral contraceptives: suppress ovulation
 - Barrier methods: block sperm
 - IUD: prevents implantation
 - Sterilization: tubal ligation or vasectomy
- Choice depends on effectiveness, side effects, and preferences

Contraception and Menopause

- **Menopause** (更年期) typically age 45-55
- Cessation of cycles for 12 consecutive months
- Ovarian follicles depleted; estrogen/progesterone decrease
- FSH and LH rise due to loss of negative feedback
- Symptoms: hot flashes, vaginal dryness, mood changes, bone loss
- **HRT** (hormone replacement therapy, 激素替代疗法) can help but has risks

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Fertilization Process

- Fertilization occurs in ampulla of uterine tube
- **Capacitation** (获能) of sperm in female tract (6-8 hours)
- Sperm must penetrate corona radiata and zona pellucida
- Only capacitated sperm can undergo acrosome reaction

Fertilization Process

- Fertilization sequence:
 - Sperm encounters corona radiata
 - Acrosome reaction releases enzymes
 - Penetration through zona pellucida
 - Fusion with oocyte membrane



Fertilization Process

- **Acrosome reaction** (顶体反应)
essential for fertilization
- Acrosome contains **hyaluronidase** (透明质酸酶) and **acrosin** (顶体酶)
- Enzymes digest path through protective layers
- Acrosomal membrane fuses with plasma membrane



Fertilization Process

- Upon fertilization, secondary oocyte completes meiosis II
- Forms second polar body
- Male and female pronuclei fuse
- **Cortical reaction** (皮质反应) prevents polyspermy
- Zona pellucida modified to block additional sperm



Fertilization Process

- **In vitro fertilization (IVF)** (体外受精) occurs outside body
- **ICSI** (intracytoplasmic sperm injection, 卵胞浆内单精子注射)
- Single sperm injected directly into oocyte
- Used for severe male infertility



Early Development

- **Cleavage** (卵裂) begins immediately after fertilization
- Rapid mitotic divisions without cell growth
- Zygote $\rightarrow$ 2 $\rightarrow$ 4 $\rightarrow$ 8 cells (day 3)
- **Morula** (桑椹胚): 16-32 cells (day 3-4)
- **Blastocyst** (囊胚): hollow ball (day 5-6)



Early Development

- Scanning electron micrographs show early stages
- 4-cell stage shows individual blastomeres
- 16-cell morula stage shows compacted cells
- These stages occur during transport through uterine tube



Early Development

- Blastocyst structure:
 - **Trophoblast** (滋养层): outer layer, forms placenta
 - **Inner cell mass** (内细胞团): becomes embryo proper
 - **Blastocoel** (囊胚腔): fluid-filled cavity
- Differentiation begins before implantation

Implantation

- **Implantation** (着床) begins day 5-7, completed by day 10
- Blastocyst adheres to endometrium
- Trophoblast invades uterine tissue
- **Syncytiotrophoblast** (合体滋养层) invades endometrium
- **Cytotrophoblast** (细胞滋养层) remains as cellular layer



Implantation

- **hCG** (human chorionic gonadotropin, 人绒毛膜促性腺激素) from trophoblast
- Maintains corpus luteum during first trimester
- Ensures continued progesterone secretion
- Detected in blood/urine - basis for pregnancy tests
- Peaks at 8-10 weeks



Placenta Formation

- Placenta develops from trophoblast and maternal tissue
- **Chorionic villi** (绒毛) are fingerlike projections
- Contain fetal blood vessels
- Increase surface area for maternal-fetal exchange



Placenta Formation

- Placental structure:
 - **Amnion** (羊膜) surrounds fetus in amniotic fluid
 - **Chorion frondosum** (绒毛膜): fetal portion
 - **Decidua basalis** (蜕膜底层): maternal portion
 - **Umbilical cord** (脐带): two arteries, one vein



Placenta Formation

- **Amniocentesis** (羊膜穿刺术) detects genetic disorders
- Performed at 15-18 weeks
- Amniotic fluid withdrawn for analysis
- Chromosomal and biochemical testing possible



Placenta Formation

- Blood circulation within placenta
- Maternal and fetal blood separated but do not mix
- Exchange occurs across placental barrier
- Fetal blood: umbilical arteries (deoxygenated) → villi → umbilical vein (oxygenated)



Placenta Formation

- Exchange across placenta by multiple mechanisms:
 - Diffusion: O_2 , CO_2 , nutrients, wastes
 - Active transport: amino acids, vitamins, minerals
 - Receptor-mediated endocytosis: IgG antibodies
- Placenta barrier protects but not absolute
- Alcohol, drugs, some viruses can cross

Placental Endocrine Function

- Placenta produces hormones requiring fetal-maternal cooperation
- Maternal cholesterol → placental progesterone
- Fetal DHEAS → placental estrogens
- Demonstrates integrated fetal-placental-maternal unit



Placental Endocrine Function

- Major placental hormones:
 - hCG: maintains corpus luteum
 - hCS: promotes growth, metabolic changes
 - Estrogens: uterine growth, breast development
 - Progesterone: maintains endometrium, inhibits contractions



Labor and Parturition

- **Parturition** (分娩) is childbirth process
- **Labor** (产程) involves uterine contractions
- Positive feedback mechanism initiates labor
- Fetal adrenal → DHEAS and cortisol
- Cortisol → placental CRH → more cortisol (positive feedback)
- Estriol, prostaglandins, oxytocin → contractions



Labor and Parturition

- Three stages of labor:
 - Stage 1: cervical dilation (8-24 hours)
 - Stage 2: delivery of baby (30 min - 2 hours)
 - Stage 3: delivery of placenta (5-30 minutes)
- **Oxytocin** (催产素) stimulates contractions
- Contractions → push fetus → stretch cervix → more oxytocin (positive feedback)

Lactation

- Mammary gland structure:
 - **Lobules** (小叶) with **alveoli** (腺泡) produce milk
 - **Lactiferous ducts** (输乳管) transport milk to nipple
- Development during pregnancy prepares for lactation



Lactation

- Hormonal control of lactation:
 - During pregnancy: estrogen/progesterone stimulate development but inhibit milk production
 - After delivery: hormone drop allows prolactin to stimulate milk production
 - High estrogen during pregnancy blocks prolactin's lactogenic effect



Lactation

- Lactation involves two processes:
 - **Milk production**: stimulated by prolactin from anterior pituitary
 - **Milk ejection**: stimulated by oxytocin from posterior pituitary
- Suckling triggers neuroendocrine reflex releasing both hormones



Lactation

- Maternal antibodies protect newborn:
 - IgG crosses placenta during pregnancy (protects 3-12 months)
 - IgA in colostrum and breast milk protects intestine
 - **Colostrum** (初乳): first milk, rich in antibodies
- Passive immunity bridges gap until infant immune system matures



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- Sexual reproduction maintains genetic diversity through meiosis and fertilization
- Sex determination by SRY gene; hormones direct sexual differentiation
- HPG axis regulates reproduction through feedback loops
- Male reproduction: continuous spermatogenesis; testosterone production
- Female reproduction: cyclic ovulation; coordinated hormonal changes
- Fertilization initiates development; implantation establishes pregnancy
- Placenta mediates maternal-fetal exchange and hormone production
- Parturition initiated by positive feedback; lactation nourishes newborn